

GURMANDER SINGH MAAN

AI/ML Engineer / Software Developer

Vancouver, BC, Canada

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TECHNICAL SKILLS

Languages	Python, JavaScript, Java, SQL, Bash
AI / ML	PyTorch, Scikit-learn, Hugging Face, RAG, Agentic AI, LLM APIs, CNNs, NumPy, Pandas, Torchaudio
Backend / Data	FastAPI, Flask, Express, REST APIs, SQLAlchemy, Pydantic, PostgreSQL, Redis, Celery
Frontend / DevOps	React, Vue.js, Docker, Git, GitHub Actions, CI/CD, Linux/Unix, CLI/Terminal, AWS

EDUCATION

Indian Institute of Technology Madras BS in Data Science and Applications Relevant Coursework: Deep Learning, Machine Learning, Software Engineering, Software Testing, Database Management Systems, AI: Search Methods.	Apr 2024 – Apr 2027 (Expected) GPA: 9.5/10
Langara College - Vancouver, BC Associate of Science	Jan 2017 – Jul 2019

PROJECTS

Syngenta Field Force Intelligence <i>IIT Madras × Syngenta Hackathon · First Runner-up · FastAPI · React · OR-Tools</i>	May 2026
- Implemented Google OR-Tools route optimization with OSRM road-distance matrices and route geometry, including geospatial and coordinate-processing workflows for multi-stop field routes. - Contributed to AI voice/route-analysis features and GitHub Actions CI/CD, including automated testing and deployment of the backend to Hugging Face and frontend to Netlify.	
Messy Mashup - Music Genre Classification <i>PyTorch · Librosa · Torchaudio · W&B</i>	Mar 2026
- Built a 10-class audio-classification pipeline for noisy, distribution-shifted data using mel-spectrograms, delta features, SpecAugment, synthetic mashups, and ESC-50 environmental-noise augmentation. - Trained and compared CNN, CNN-Transformer, and Audio Spectrogram Transformer architectures; tracked experiments with W&B and achieved 0.945 Macro F1, ranking 86th of 1,119 participants (top 8%).	
Urban Fix - AI-Powered Civic Service Platform <i>FastAPI · PostgreSQL · Vue.js · YOLOv11 · Gemini Vision · Docker</i>	Jun 2026 – Aug 2026
- Developed backend and AI workflows for role-based civic complaints, including geolocation-based ward assignment, lifecycle tracking, and photo-based verification using FastAPI, SQLAlchemy, PostgreSQL, and JWT authentication. - Integrated YOLOv11 for pothole/garbage detection and vision models for broader issue analysis; added GitHub Actions CI for automated unit/integration tests and Ruff linting on pushes and pull requests.	

EXPERIENCE

TELUS <i>SmartHome Security Customer Support</i>	Jan 2022 – Jan 2024 Surrey, BC
Resolved device, connectivity, and setup issues for residential SmartHome Security systems, translating technical diagnostics into clear customer guidance in a customer-facing support environment.	
TELUS <i>Install & Repair Technician</i>	2020 – 2021 Surrey, BC
Installed and troubleshooted residential internet/Wi-Fi, television, and phone services onsite, diagnosing network and hardware issues and completing service work independently.	

Achievements

IIT Madras Academic Distinction — CGPA ≥ 9.5 at end of Foundation and Diploma Levels	2026
IIT Madras × Syngenta Hackathon — First Runner-up	2026
Dean's Honour Roll, Langara College	Spring 2018
AWS Certified Cloud Practitioner	May 2023 – Apr 2026